

Due: October,13, 11:59 PM, submitted on canvas, PDF only.

This document is meant to serve as a roadmap for your thesis. It does not need to be a polished document, but it *does* need to state a clear and compelling argument as to what your scientific contribution is to the field and your plan to execute it within the honors program timeline. At this time, we want to prioritize logic and feasibility of your ideas over specific figures or equations. Simple sketches or descriptions are sufficient to illustrate your methodology and explain outcomes. Do not spend time creating professional figures at this point.

Our expectation is for you to clearly articulate the problem you are trying to solve, state the gap in the literature, and present your plan to complete the research. While the exact format is up to you and your advisor, make sure you include a timeline to clearly demonstrate the feasibility of the research you are proposing. You also need to include a section titled "Honors Thesis Declaration", which is included in point 5 below:

Suggested Format:

1. Background and Significance: What is the motivation behind your project and the impact it has in your field or other engineering applications?

2. Research Goals/Hypothesis: Clearly define the tasks that you have set forth for the project and/or the hypothesis you are investigating, including what has already been accomplished, and what you're planning to do before the end of the project.

3. Proposed Methods: How are you going to accomplish these goals?

4. Preliminary Results/Expected Outcomes: If you have preliminary data, show it here. Otherwise, state what the expected results and outcomes of the project are.

5. Honors Thesis Declaration:** In this section, you need to carefully distinguish this research from any other required coursework, design project, or capstone project that is required for your concentration. Overlap between such projects is allowed, but you need to distinguish and clarify how this honors project has gone above and beyond the required curriculum. If there is no overlap with any other coursework, you may state so here.

Abstract:

Minimally invasive surgery (MIS) utilizes small incisions to place a camera on a rod - called a laparoscope - and tools within a patient so that a surgeon can operate via video feed. Although MIS provides many benefits for patients' recovery and safety, there are currently some shortcomings in the field of view and depth perception available via laparoscopy.

This project proposes the design and manufacture of a prototype to overcome these shortcomings. The prototype will consist of an angled laparoscope that can rotate 360 degrees during an operation to allow surgeons to have multiple views of the surgical site.

This goal will require: 1) designing, manufacturing, and testing hardware that may be safely placed within a patient and capture necessary data, 2) development of computer vision software so that the images captured may be composed through three dimensional scene reconstruction into a high resolution image with depth perception, and 3)) ex vivo testing to evaluate the performance of the device.

Through the combination of mechanical design and computer vision algorithms, this project will yield a device that improves the efficacy of minimally invasive surgery.

Background and Significance:

Laparoscopy is a type of minimally invasive surgery that allows surgeons to treat abdominal disorders. The general process of laparoscopic surgery includes 1) small incisions within the abdomen; 2) inflation of the abdomen through the use of CO₂ to create more space for surgical instruments; 3) The insertion of a camera on a small rod of approximately 5-10 mm in diameter called a laparoscope to allow surgeons to see their work as they operate; 4) Operation upon the patient using specialized instruments. This surgical technique is used for the diagnosis and treatment of appendicitis, gastrointestinal reflux, hernias, some types of cancer, and other diseases related to organs in the abdomen (Buia et al., 2015).

Some of the benefits of Laparoscopic surgery include "a speedy recovery, shorter duration of hospital stay, minimal postoperative pain, discomfort and disabilities, and better cosmetic outcomes"(Basunbul et al., 2022). These benefits make it the preferred treatment method for many conditions and a revolutionary advancement in the field of surgical methods. However, this technique does not come without its drawbacks, particularly within the logistical difficulties of surgeons to use this inherently limiting device. These drawbacks include ergonomic difficulties for surgeons using the device, high training investment to become adept at the precise movements necessary to perform operations, and fatigue during operation (Basunbul et al., 2022). One particular point of difficulty is the limited field of view and constant need of adjustments by a camera operating assistant to maintain visuals on the operating area (Rahman & Mishra, 2011). In addition, the camera lens may be impaired or fogged for up to 37% of

operation time according to Yong et. al. (2015). Fogged lenses can result in removal and reinsertion of the laparoscope that could be potentially risky and increase surgical times.

Increasing the field of view of the laparoscopic camera would be a major step toward addressing these limitations. Typical laparoscopes have a limited field of view of approximately 70 degrees that allows for focused viewing of vital regions of the surgery, but provides a visual comparable to looking through a keyhole on a door. This limited view can result in injuries on the periphery of the surgery. By developing systems to address this, the safety, time, and effectiveness of laparoscopic surgery can be improved, making these operations more likely to succeed.

A few different systems have been developed in the past to potentially address these pitfalls. Shapira et. al. (2023) describes SurroundScope, a system that increases the view of the laparoscope from 70 degrees to 270 degrees and was successfully used in 21 Cholecystectomies and was reported to increase the safety, reduce the time, and improve planning of surgeries. However, the high cost of equipment and requirement for new training has prevented this technology from going into widespread use. Other approaches include: a dual-camera device that utilizes optical beam splitting and computer processing to pan and zoom digitally through a doubled 140 degree field of view (Katz et. al, 2022); a magnetically adjustable camera system with 150 degree adjustable view, enhanced lighting and lens cleaning micro-pump (Yazdanpanah et. al., 2025); and the use of shape memory alloy actuation to allow up to 178 degree adjustment of the camera (Pietsch et. al., 2025). However, each of these approaches respectively face challenges of bulkiness and difficulty of use, sizing of the system to be appropriate for surgery, and temperature regulation.

Research Goals/Hypothesis:

Goals: The goal of this project is to create a novel laparoscope system that meets the following criteria:

1. Provide a 180-270 degree field of view for surgeons so that they can have maximum visual information as they perform operations
2. Integration of existing and widely used laparoscopes to reduce the cost of this novel technology and barriers to entry for surgeons to learn on the novel system
3. Consideration of the safety, computational, and size constraints necessary for this system to be fully integrated into laparoscopic operations.

Hypothesis: The rotation of an angled laparoscope can capture many images that can be stitched together using computer vision techniques to provide a complete view of the surgical scene with high resolution and depth perception. This system will overcome the pitfalls of existing laparoscopes while having lower barriers to entry than other approaches.

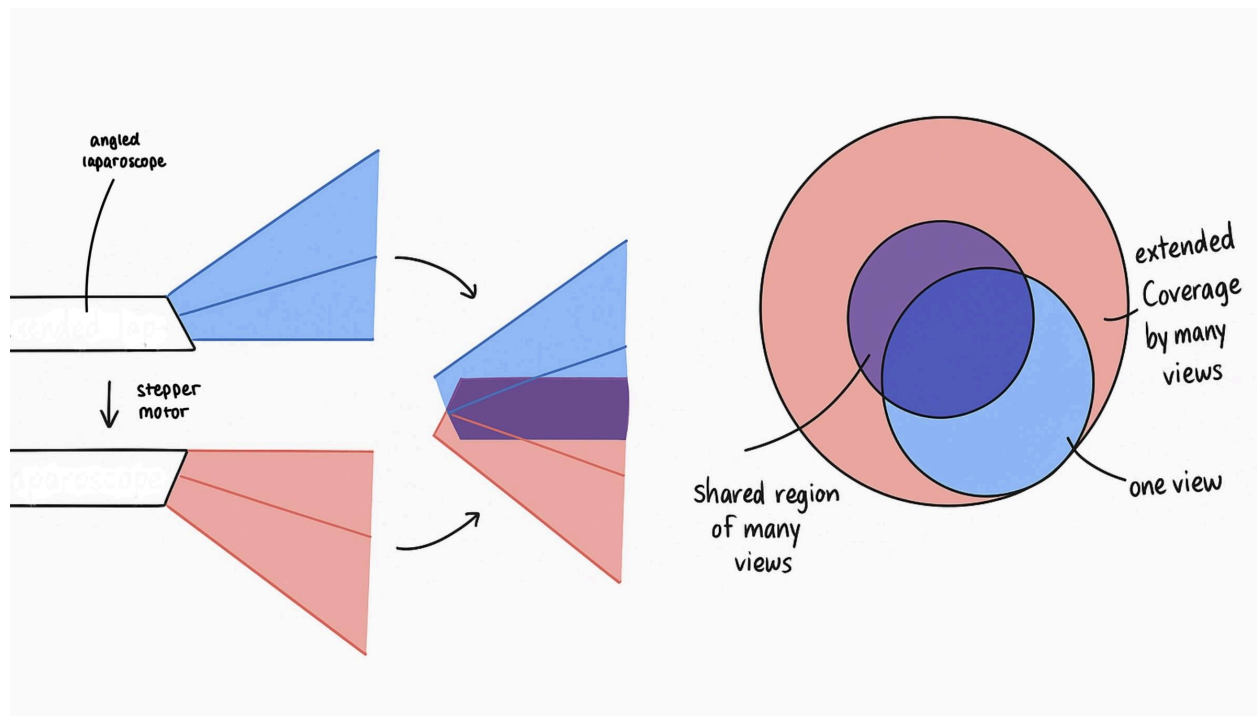
Current Progress: Investigation of manufacturing techniques, mechanical design, and electronics of such a system is already underway. The finished product of this project will include a large-scaled proof of concept that demonstrates the mechanical principles of the project, a computer vision algorithm to reconstruct the scene from many consecutive images by the rotating laparoscope, and a design for surgical use that utilizes an existing laparoscopic device.

Proposed Methods:

This project will involve iterative prototyping of the subsystems required to make the rotating laparoscope system. System components would include:

- A stepper motor to drive the rotation of the camera 360 degrees about its central axis and stop at regular intervals to capture images.
- A housing for the laparoscope with integrated ball bearings to allow for continuous rotation while protecting the patient, not obscuring the lens of the camera, and staying within the diameter necessary to allow for insertion into micro-incisions.
- Rotating electronics connections that allow the camera to continue transmitting data while achieving the desired range of motion
- Communication with a computer that continuously runs computer vision software in order to run computer vision algorithms such as odometry to generate full views of the surgical site.

A diagram of the concept for the overall proposed system is shown in Figure 1:



Methods to achieve this goal will include 3D printing, Computer Aided Design, programming in OpenCV with Python for image processing, and design and testing of embedded systems using Arduino.

Expected Outcomes:

The initial goal will be to design and manufacture a 3D printed model of a larger scaled version of the system, utilizing arduino and a larger camera with comparable parameters to begin the development of computer vision odometry algorithms. Then, testing will proceed on a system that integrates an existing laparoscope with a final goal of simulated surgery to test effectiveness. Parameters of note to test will include field of view, image quality, update rate, size of system, and stability of system.

Current progress has included comparative analysis of existing laparoscopic systems (Table 1) and identifying key parameters needed to implement the novel rotating system (Table 2).

Table 1: Comparison of Laparoscopic Camera Systems Available

Product	Company	Hyperlink	Datasheet	Type	Field of Vi	Cabling	Lights	Angle	Diameter Opto	Length Opt	Shutter Type	Frame Rate	Shutter Speed (Range)	Notes
Looking Glass® 4K	CONMED	https://www.conmed.com	https://www.conmed.com	3 chip CMOS, integrated Stand		USB	no spec, eLuma™, fiber optic	no info	no info	no info				2x zoom , 4k
Laparoscope	CONMED	https://www.conmed.com		Camera Head				0, 30, 45	5.5mm, 10mm	no info				
Rigid Laparoscopes	Olympus	https://medical.olympus.com		Camera Head, aspherical anti-distortion lens				0, 30, 45	5mm, 10mm	430mm				
Autoclavable Camera	Olympus	https://medical.olympus.com		Camera										Focal distance f = 15.8 – 31.3 mm, 1.2-1.5x zoom, 1080p high depth of field, 16:9 wide-screen, fog free
Video Laparoscope, ENDOEYE II	Olympus	https://medical.olympus.com		CMOS or CCD	85°			0, 30	5mm, 10mm	no info		50-60 Hz		
1588 AIM laparoscopes	Stryker	https://www.stryker.com		Camera Head					5.4mm					
1588 AIM Camera	Stryker	https://www.stryker.com	https://www.manual.stryker.com	Progressive Scan CMOS	30°-70°							60Hz	1/60 – 1/50,000 second	9 modes
Info Page	Medtronic	https://www.medtronic.com												
TipVision	Medtronic	https://www.medtronic.com	https://www.medtronic.com	Integrated Chip Scope			integrated LEDs, 30,000 hour lifespan	0, 30, 45	5mm, 10mm	344mm, 312mm				depth of field 20-200mm
Storz Image 1 Laparoscope	Storz	https://www.endoscopy.com		CMOS Progressive Scan	70°				0					

Table 2: Notable Parameters for Proposed Laparoscopic System

Parameter	Diameter (mm)	Field of View (Degrees)	Shutter Type	Image Refresh Rate
Existing Systems	5-10	~70	Global or Rolling	60 fps
Requirement for Rotating System	<10	~180	Global	~40 fps
Explanation	System needs to fit within micro-incisions	Maximum field of view, achieved via rotation	Allows instant image capture without interference from rotation	~20 images per revolution * 2 rotations per second = ~40 fps

Honors Thesis Declaration:

This project has no overlapping components with my coursework or capstone project. My capstone project is focused on the development of an interactive display table for telecommunication. I have no prior coursework or research related to medical devices. This project will proceed entirely independently of my other endeavors.

Timeline:

October	Design + initial 3D printed prototyping, electronics development
November	Implementation of CV algorithm on 3D printed prototype, begin integration with existing laparoscope
December	Comprehensive testing + data collection on prototypes, see if key parameters achieved
January	Continued testing + experimental analysis
February	Thesis Drafting
March	Thesis Drafting
April	Thesis Submission

Sources:

Basunbul LI, Alhazmi LSS, Almughamisi SA, Aljuaid NM, Rizk H, Moshref R. Recent Technical Developments in the Field of Laparoscopic Surgery: A Literature Review. Cureus. 2022 Feb 15;14(2):e22246. doi: 10.7759/cureus.22246. PMID: 35340455; PMCID: PMC8929761.

Buia A, Stockhausen F, Hanisch E. Laparoscopic surgery: A qualified systematic review. World J Methodol. 2015 Dec 26;5(4):238-54. doi: 10.5662/wjm.v5.i4.238. PMID: 26713285; PMCID: PMC4686422.

N. Yong, P. Grange, and D. Eldred-Evans, "Impact of laparoscopic lens contamination in operating theaters: A study on the frequency and duration of lens contamination and commonly utilized techniques to maintain clear vision," Surgical Laparoscopy Endoscopy & Percu-taneous Techniques, vol. 26, no. 4, pp. 286–289, 2016.

Rahman, Muhammad Nurhadi & Mishra, Rajneesh. (2011). The Camera-holding Robotic Device in Laparoscopy Surgery. *World Journal of Laparoscopic Surgery with DVD*. 4. 132-135. 10.5005/jp-journals-10007-1130.

Shapira, S.S., Ehrlich, Z., Koren, P. *et al.* Comparing a novel wide field of view laparoscope with conventional laparoscope while performing laparoscopic cholecystectomy. *Surg Endosc* 37, 8910–8918 (2023). <https://doi.org/10.1007/s00464-023-10393-3>

Pietsch, M., Henkel, K., Körbitz, R. *et al.* A deflectable video camera system for laparoscopic surgery based on shape memory alloy actuators. *Sci Rep* 15, 11354 (2025). <https://doi.org/10.1038/s41598-025-95898-8>

A. R. Yazdanpanah, X. Liu, N. Li and J. Tan, "A novel laparoscopic camera robot with in-vivo lens cleaning and debris prevention modules," 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Vancouver, BC, Canada, 2017, pp. 3669-3674, doi: 10.1109/IROS.2017.8206212.